PharmAchieve

EVIDENCE-BASED MEDICINE, STUDY DESIGN & BIOSTATISTICS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care:
Clinical Care

Providing Care:
Distribution

**Knowledge
& Expertise**

Communication
& Collaboration

Leadership &
Stewardship

Professionalism

LEGEND

ARR	Absolute Risk Reduction
ARI	Absolute Risk Increase
AR	Attributable Risk
AF	Attributable Fraction
CER	Control Event Rate
CI	Confidence Interval
EBM	Evidence-Based Medicine
EER	Experimental Event Rate
FN	False Negative
FP	False Positive
ITT	Intention-To-Treat
LOCF	Last Observation Carried Forward
NNH	Number Needed to Harm
NNT	Number Needed to Treat
NPV	Negative Predictive Value
OR	Odds Ratio
PAF	Population Attributable Fraction
PAR	Population Attributable Risk
PP	Per-Protocol
PPV	Positive Predictive Value
RCT	Randomized Control Trial
RR	Relative Risk
RRR	Relative Risk Reduction
TN	True Negative
TP	True Positive

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EVIDENCE-BASED MEDICINE (EBM)

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INFORMATION VS EVIDENCE

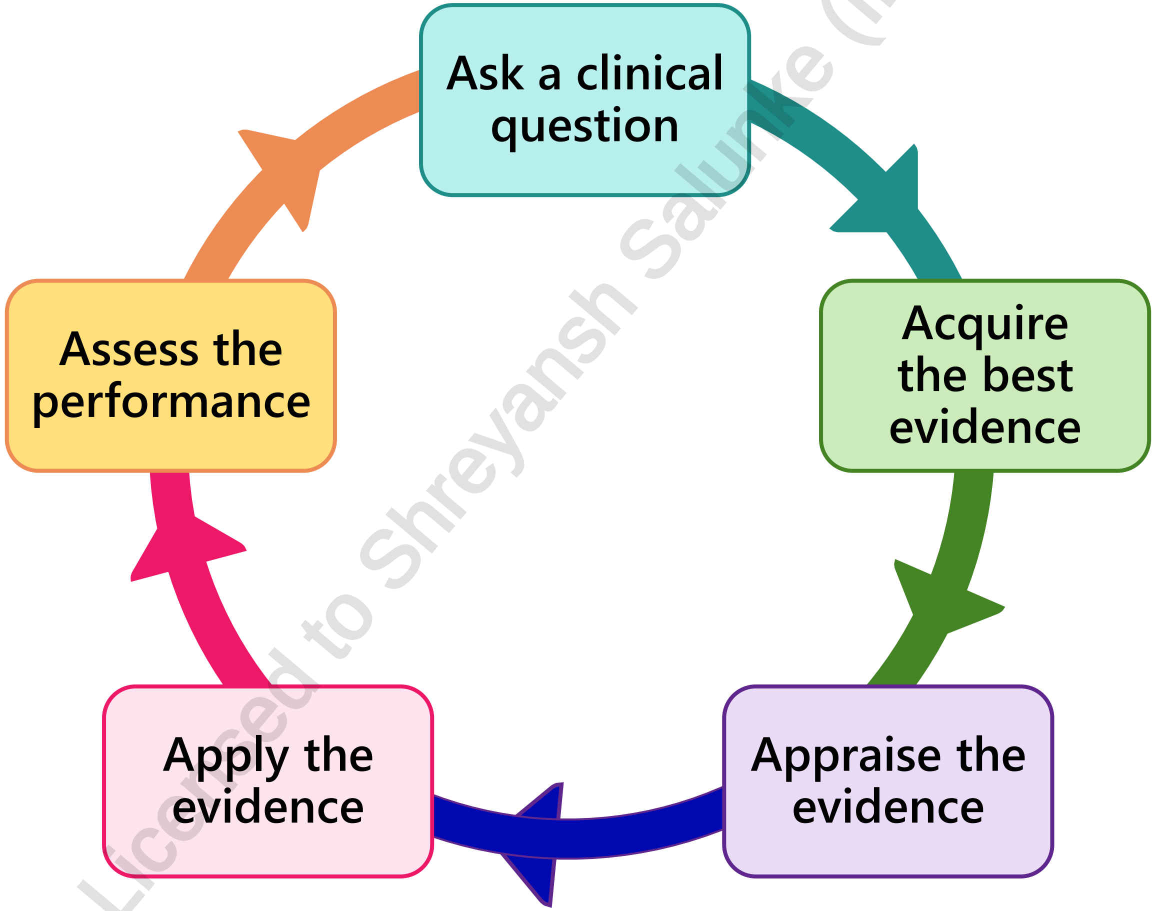
Information ≠ evidence
While all evidence is also information, not all information is evidence

- Information: also known as data; collection of facts
  - Before judgment: need to review information

- Evidence: organized data that is understood to have significance/meaning
  - After judgment: subset of data used to support a claim

EVIDENCE-BASED MEDICINE (EBM)

Uses the scientific method to organize and apply current evidence to improve healthcare decisions



ASKING A "CLINICAL QUESTION"

- It is helpful to narrow down the type of clinical question one is asking to help focus on evidence that is directly relevant to patients' clinical needs

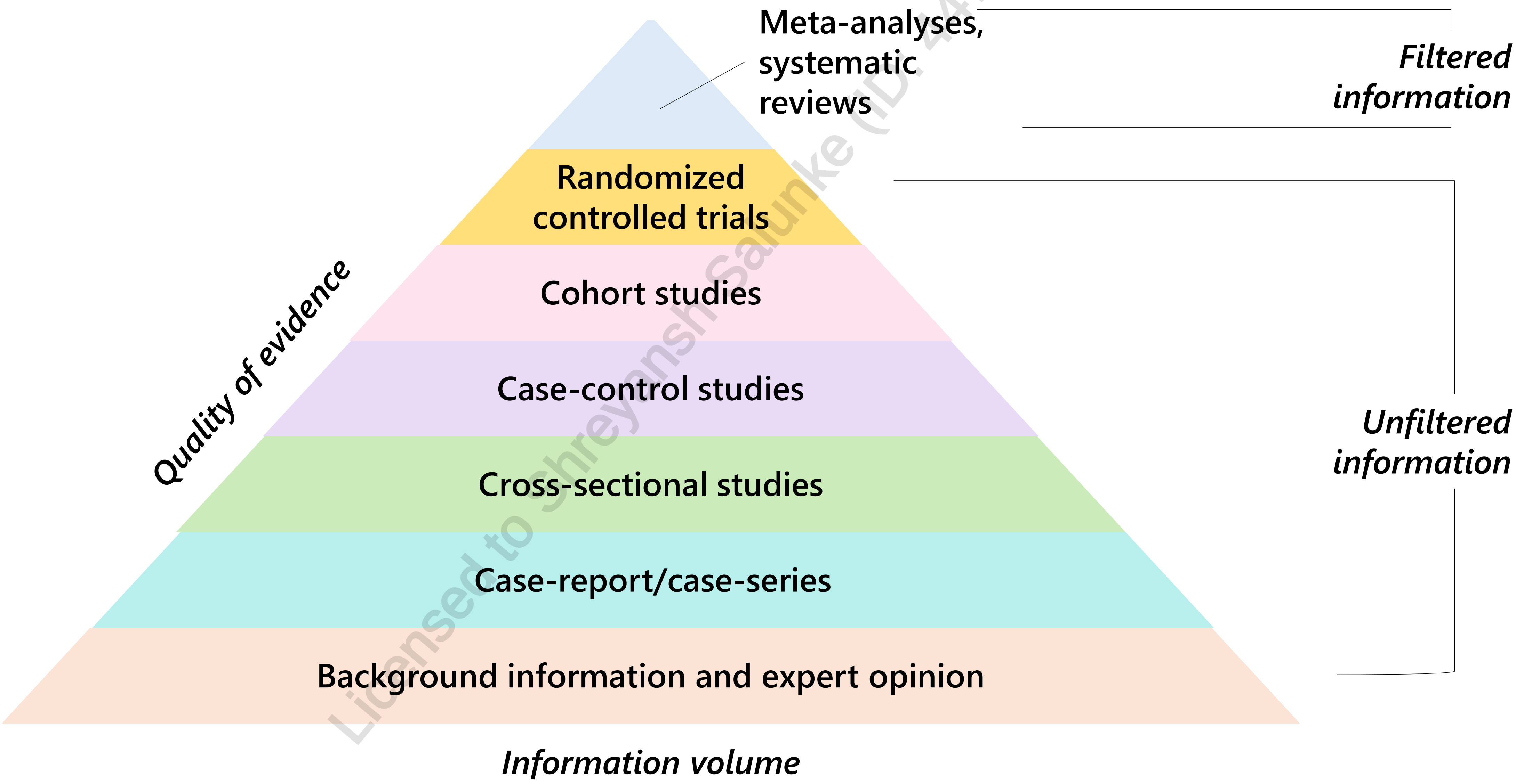
- Clinical questions have the following components:
  - P: Population, patient situation or problem of interest
  - I: Intervention
  - C: Comparison or Control
  - O: Outcome(s) of interest
  - S: Setting or Study Design

ASKING A "CLINICAL QUESTION"

What is the efficacy of metformin versus rosiglitazone on glycemic control for children with type 2 diabetes in a hospital setting?

https://mdanderson.libguides.com/c.php?g=249812&p=1663282

ACQUIRING THE "BEST" EVIDENCE



APPRAISING EVIDENCE

- Evidence from clinical trials is fundamental to ethical medical practice and central to effective clinical decision-making

- Questions to help interpret study results:
  - Do the trial results reflect true effects of intervention, rather than artifactual ones (validity)?
  - Do the results suggest that the intervention is clinically useful (importance)?
  - Could the results apply to individual patients encountered in daily practice (relevance)?

CRITICAL APPRAISAL: INTERNAL VALIDITY

Internal validity: measure of a study's quality. How accurately do its results reflect the studied group?

- Results attributed to interventions? Comparable participants at baseline?
- Adequate statistical power (sample size)?
- Randomization?
- Blinding?
- Adequate study duration?
- Low/balanced drop-out rate and reason for drop-outs clearly stated?
- Objective, measurable outcomes?
- Reported p-values and confidence intervals?
- Evidence of bias?

If internal validity is not met, then applicability (external validity) is irrelevant

RANDOMIZATION

- Randomization: balanced process of allocating patients equally to each group
- Each patient has an equal chance of being assigned to any group
- Ideally, allocation occurs via computer-generated numbers (sequence generation)
  - Researchers are not involved in the allocation process
  - The allocation process is not easy to predict
- Randomization is done to prevent selection bias in assigning patients to a particular group
- Allocation concealment prevents researchers from influencing which participants are assigned to a given group

BLINDING

Blinding is a procedure in which one or more parties are kept unaware of the group allocation of participants.

Type	Description
Unblinded or open label	All parties are aware of group allocation
Single-blind or single-masked	One party (either the participants or clinicians/researchers) is unaware of group allocation
Double-blind or double-masked	The participants AND the clinicians/researchers are unaware of group allocation
Triple-blind	Participants, clinicians/data collectors and outcome adjudicators/data analysts are ALL unaware of group allocation

TYPES OF BIAS

Type of bias	Description
Randomization bias	When the process for randomization is, in fact, not completely random
Selection bias	When subjects studied are not representative of the target population about which conclusions are to be drawn
Performance bias	A systematic difference in the care provided to participants depending on their allocation
Attrition bias	A difference between groups due to drop-out rates (ex. more participants from the intervention group drop out than from the comparison group)
Detection bias	When groups are assessed for outcomes differently by the data analysts/outcome adjudicators

TYPES OF BIAS

Type of bias	Description
Publication bias	Studies where there is a positive finding are more likely to be published than studies in which there is a negative finding
Language bias	A tendency for publications to be published in the English language first
Healthy user bias	The concept (especially in observational studies) that individuals who take certain medications/supplements are inherently more health conscious and thus healthier
Recall bias	The concept (especially in observational studies) that individuals who experienced a negative outcome are more likely to recall exposures than individuals who did not
Volunteer bias	The concept (especially in experimental studies) that individuals who would volunteer for a study are inherently different than those who would not

CONFLICTS OF INTEREST

- Conflicts of interest: circumstances in which a professional's judgment or action regarding their primary interest is at risk of influence by a secondary interest.
  - Secondary interest could be financial or non-financial
  - Results in conscious or unconscious bias

- Secondary interest (usually financial) can influence the research to obtain and present results in favour of the secondary interest.
  - There is evidence of an association between industry sponsorship of drug trials and more favourable trial results.

CRITICALLY APPRAISE AVAILABLE EVIDENCE: EXTERNAL VALIDITY

- External validity: applicability of evidence to your patient/practice
    - Statistical significance ≠ clinical significance
        - Statistical significance: the math. Are the results a fluke? Could the difference between interventions simply be due to chance?
        - Clinical significance: Would applying the results of this study truly make a difference for your patient(s)?
    - Generalizability of results (inclusion/exclusion criteria)?
    - Applicability to real-world practice?

STATISTICAL INTERPRETATION

Statistical significance	Clinical significance
Ruled by the p-value ($p < 0.05$) If a difference is statistically significant, it simply means it was unlikely to have occurred by chance	If a treatment makes a positive and noticeable improvement to a patient, we can call this 'clinically significant'

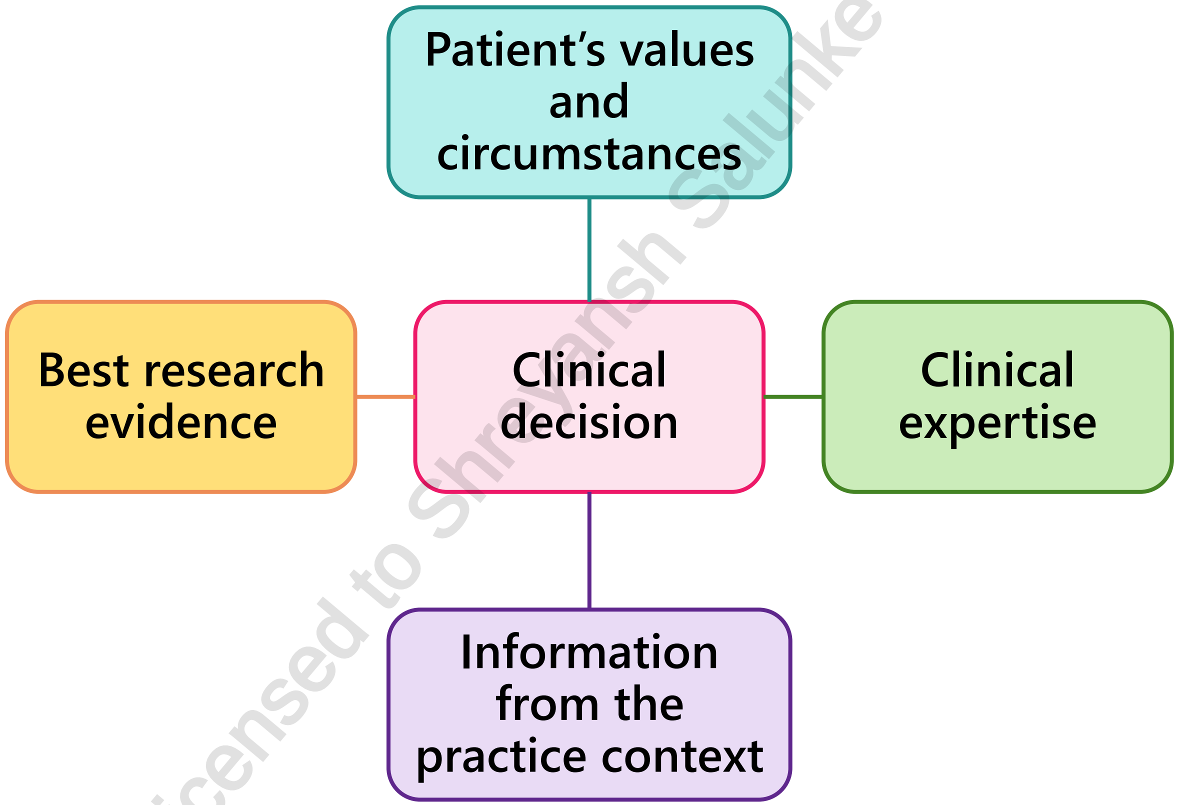
Example: study testing overall survival in patients with advanced pancreatic cancer given gemcitabine (control) or gemcitabine + erlotinib (intervention).

Average survival → found to be significantly higher in intervention group (6.24 months vs 5.91 months; $p = 0.038$)
Actual time difference → only 10 days
→ not clinically significant to most oncologists (short time, added toxicity, expensive)

https://s4be.cochrane.org/blog/2017/03/23/statistical-significance-vs-clinical-significance/

APPLYING EVIDENCE

If you have found evidence that is valid, significant and generalizable to your patient, you must decide whether and how to apply the findings to their care.



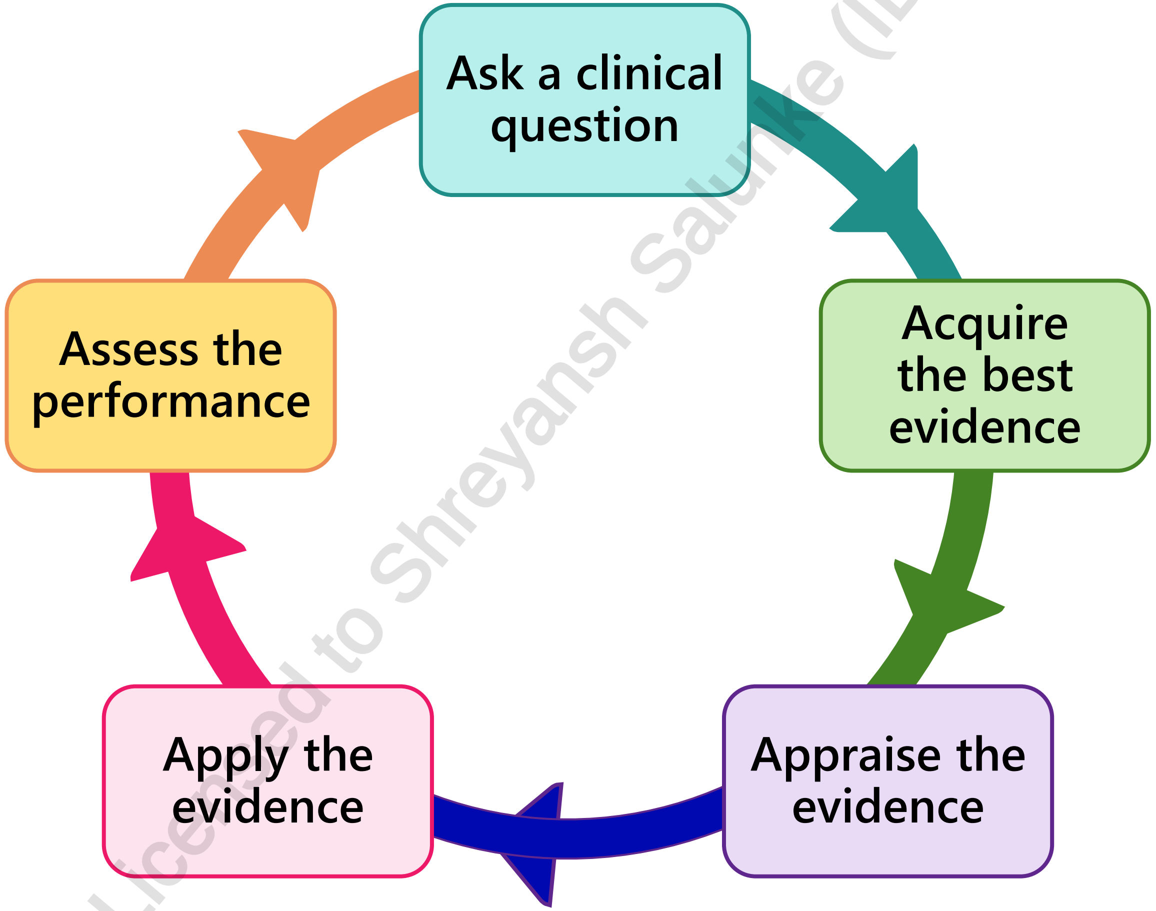
ASSESSING PERFORMANCE

- Without continuous re-evaluation, the medical provider will not be sure if their impact is positive or negative

- Self-evaluation:
    - Did you ask an objective/measurable clinical question (PICOS)?
    - Did you find the best possible evidence?
    - Did you critically appraise the evidence and evaluate it for its validity and potential usefulness?
    - Did you integrate critical appraisal of the best available evidence with individual clinical expertise in daily practice?

SUMMARY

EBM starts with the clinical question and returns to the clinical question to see whether the process worked



RESEARCH ARTICLE STRUCTURE

TITLE, ABSTRACT

↓

INTRO, OBJECTIVES

↓

METHODS (STUDY DESIGN)

↓

RESULTS, STATISTICAL ANALYSIS

↓

DISCUSSION, CONCLUSION

↓

REFERENCES, FUNDING SOURCES

THE COCHRANE COLLABORATION

An international, not–for–profit organization that aims to help people make well–informed decisions about healthcare by preparing, maintaining and promoting the accessibility of systematic reviews of the effects of healthcare interventions

- The group performs systematic reviews and meta-analyses and publishes them to the Cochrane Library for public access
- Reviewers in 43 different countries that contribute to the gathering, analysis and publication of medical research findings
- Publications produced by Cochrane are considered the most accurate and up-to-date evidence available for EBM
- Other organizations include: Joanna Briggs Institute, Centre for Reviews and Dissemination, EPPI-Centre, Center for Evidence-Based Management

RESEARCH ETHICS

All Clinical Studies must be approved by a research ethics board/ethics committee

- Research ethics: specific set of principles that researchers must demonstrate they will follow
- Primary goal → ensure study participants are protected (that they will not be harmed as a result of participation in the study) and research is conducted to serve the need of participants and society
  - Ensure data integrity and study compliance

- The Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2) provides ethics guidance that applies to all research involving human participants conducted under the auspices of an institution eligible for funding by the federal Agencies (CIHR, NSERC, SSHRC)
  - CIHR → The Canadian Institutes of Health Research
  - NSERC → The Natural Sciences and Engineering Research Council of Canada
  - SSHRC → The Social Sciences and Humanities Research Council of Canada

STUDY DESIGN

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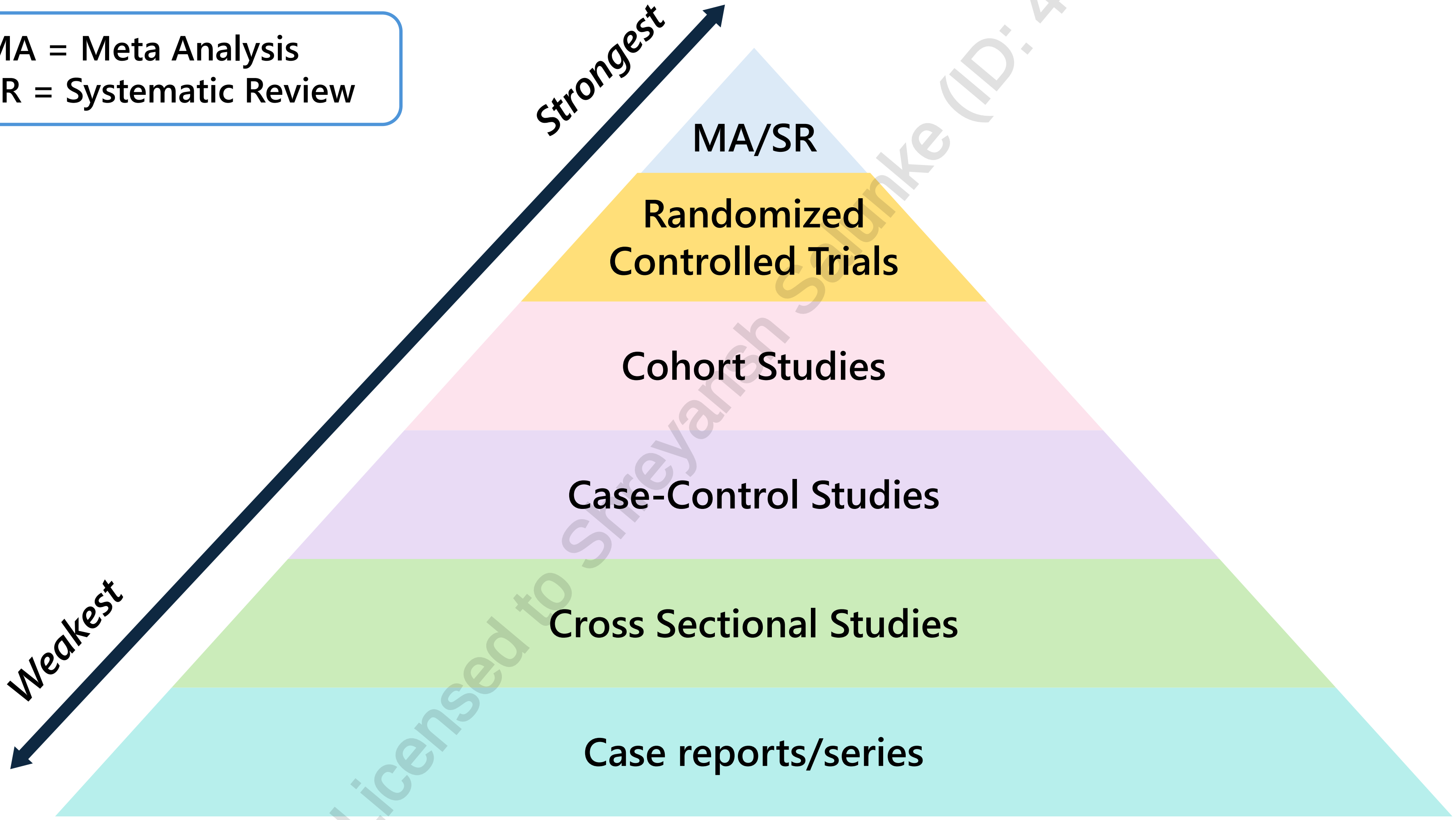
STUDY DESIGNS: OVERVIEW

- Experimental studies: compare the effect of treatments/interventions with a control or another treatment in humans
- Observational studies: non-intervention and non-experimental; factors are not controlled and subjects are simply observed
- Economic evaluations: measuring the cost of potential intervention
  - Includes cost analysis, cost-minimization, cost-utility, cost-effectiveness, cost-benefit
- Meta analyses/Systematic reviews: the review of multiple studies which address the same topic

STUDY DESIGNS- HIERARCHY OF EVIDENCE

MA = Meta Analysis
SR = Systematic Review



PRACTICE GUIDELINES

- Practice guidelines: statement produced by a panel of experts indicating best practices
  - Statement can also be produced by an organization after an extensive review of the literature

Advantages	Disadvantages
Created by panel of experts	Not available for every topic
Produced based on review of published literature	Can be outdated; slow to develop and update
Experience + evidence integrated for clinical decision algorithms	Expensive and time-consuming

OBSERVATIONAL STUDIES

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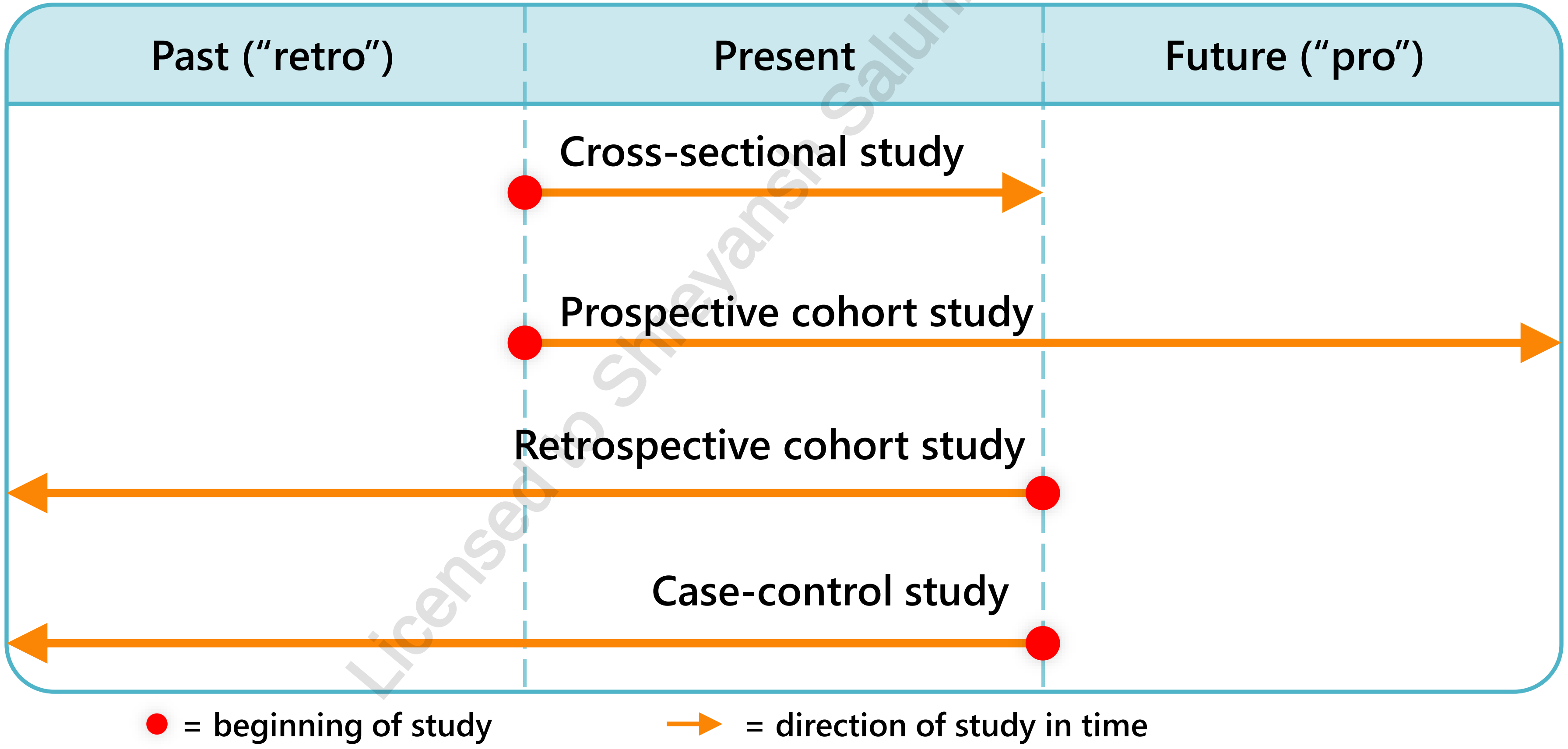
COMPARATIVE OBSERVATIONAL STUDIES

- Investigator does not intervene in the care of the patient, but simply records what happens
- Major types: cross-sectional studies, case-control studies, and cohort studies (both retrospective and prospective)



CASE REPORTS AND CASE SERIES STUDY

- Case report: documentation of a single medical occurrence, usually a rare disease or idiosyncratic reaction
  - Multiple case reports of the same occurrence is a case series
- Observational, descriptive, retrospective

Advantages	Disadvantages
Detects rare adverse effects of drugs	Difficult to compare different cases
Detects potential off-label use	Cases may not be generalizable
Cost-friendly and time-efficient	Unknown future outcome/follow-up

CROSS-SECTIONAL STUDY

- Cross-sectional study: examine relationship between disease/outcome and other variables of interest at the present time
  - Exposure and outcomes are both measured at the same time

Advantages	Disadvantages
Useful to quantify prevalence of a disease or risk factor	Can establish clear correlations but they cannot explain what caused those correlations
Describes prevalence but not incidence	Reliance on self-reported measures
Cheap and ethically safe	Establishes association at most but not causality

PREVALENCE AND INCIDENCE

- **Prevalence:** proportion of people in a population who have a particular disease at a specific point in time or over a specified period of time.
  - Expressed as a fraction, percentage or number of cases per thousand.

- **Incidence:** proportion of new occurrences of a factor over a period of time.
  - Incidence proportion: proportion of initially disease-free that develop the disease.

$$\text{Prevalence Rate} = \frac{All\ \boldsymbol{new}\ \&\ \boldsymbol{pre-existing}\ cases\ of\ a\ specific\ disease\ during\ a\ given\ time\ period}{Total\ population\ during\ the\ same\ time\ period} \times 100\%$$

$$\text{Incidence Rate (IR)} = \frac{All\ \boldsymbol{new}\ cases\ of\ a\ specific\ disease\ during\ a\ given\ time\ period}{Total\ population\ during\ the\ same\ time\ period} \times 100\%$$

Example: In a town of 10,000 people,

- On January 1st 2025, there are 200 people with diabetes → Prevalence = 200 ÷ 10,000 = 2%
- From Jan 1st – Dec 31st 2025, 50 new cases are diagnosed → Incidence = 50 ÷ 9,800 = 0.51%

CASE-CONTROL STUDIES

- Case-control study: Start with 2 groups which are defined by the outcome. The ‘case’ group experienced the outcome, whereas the ‘control’ group did not.
  - Retrospective study → once groups are defined, go back in time (i.e. through medical records or patient recall) to assess for exposure

Advantages	Disadvantages
Multiple exposures or risk factors can be examined	Quality of data may be compromised
Quick to complete as it is retrospective and no follow-up is needed	The study depends upon the history given by the subject. Hence recall bias can affect results
Easy to conduct as no follow-up is required	Finding an appropriate control may be challenging

COHORT STUDIES

- Cohort study: start with 2 groups, defined by the exposure. The ‘experimental’ group were exposed; the ‘control’ group were not. Can be prospective or retrospective.
  - Prospective cohort study → both groups are followed forward in time (i.e. for weeks, months or even years) to determine outcome
  - Retrospective cohort study → once groups are established based on exposure, existing medical records or patient recall is reviewed to assess for the outcome

Advantages	Disadvantages
Calculate rates of disease in exposed and unexposed individuals over time (e.g. incidence, relative risk)	Long-term follow-up
Useful for uncommon exposures/common diseases	Blinding/masking difficult; no randomization
Examine multiple outcomes for a given exposure	Expensive + time consuming

EXPERIMENTAL STUDIES

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RANDOMIZED CONTROLLED STUDY

- Randomized controlled study: subjects who meet the inclusion criteria are randomly assigned to an intervention or control group, considered the gold standard for determining causality
  - Prospective, interventional, experimental

Advantages	Disadvantages
Randomization minimizes bias	Time consuming and costly
Allows researchers to have control of the exposure event	Limited applicability: inclusion/exclusion criteria
Results are analyzed using statistical tools	Volunteer bias

CROSSOVER STUDY

- Crossover study: controlled trial where each study subject is randomized to both study arms sequentially with an adequate washout period
  - Each study subject acts as a control

Advantages	Disadvantages
Less participants required	Requires longer time commitment due to washout period– drop-out rate thus increases
Subject acting as own control eliminates randomization bias	Assumption of no carryover effects (from washout period) is difficult to guarantee

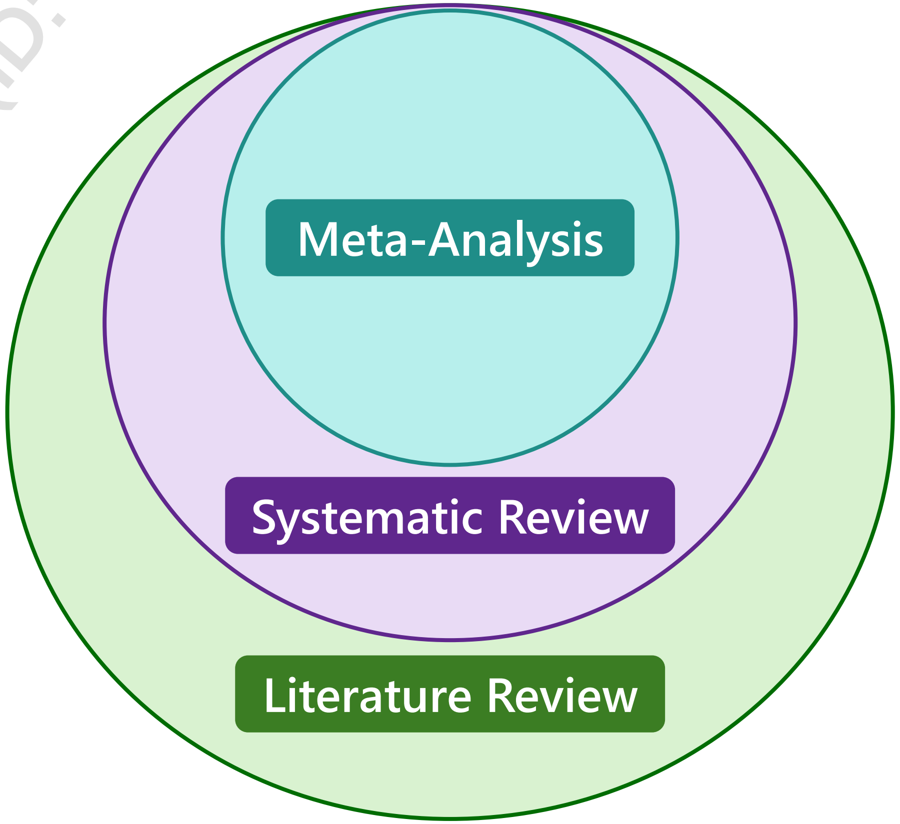
FACTORIAL DESIGN

- **Factorial experiment:** an experiment with two or more factors with discrete levels; every experimental unit takes on all possible combinations of each level
  - Tests the interactions between different factors and their effects on the outcome
  - Example of factors: gender (male/female), diet, age, socioeconomic status

Advantages	Disadvantages
Ability to investigate multiple factors	More than two factors is complicated
Considerable saving of the experimental resources, experimental material and time	Single error jeopardizes entire study

REVIEW

Literature Review	• Summarizes a topic broad in scope (e.g. cancer treatment) • Looks at everything; not specific in types of studies chosen
Systematic Review	• Answers a specific clinical question (i.e. PICOS) • e.g. Is vitamin C or chemotherapy better for cancer? • Defines a specific search strategy
Meta-Analysis	• Combines similar studies and pulls data to get statistically significant results



PER-PROTOCOL ANALYSIS

- Per-protocol analysis: analysis that only includes patients who completed the trial according to the protocol in the statistical analysis and results
  - Can overestimate effectiveness if ITT analysis is not done
  - If per-protocol analysis is presented in study, ITT should also be presented → recommended to have both included
  - Per-protocol is much more controlled analysis → not as applicable to "real" scenarios

INTENTION-TO-TREAT (ITT)

- Intention-to-treat analysis: includes the results of all randomized patients in the statistical analysis regardless of their adherence to inclusion criteria, treatment received, drop-outs or loss to follow-up
  - Patients who withdrew from the trial or who were lost to follow-up are counted as a negative outcome or last observation carried forward (LOCF) method where the last measurement available is used for analysis
  - Need to include ITT data since withdrawal occurring after randomization may be due to treatment received
  - ITT should be primary analysis method of most clinical trials

AS-TREATED ANALYSIS

- As-treated analysis: analysis that only includes patients who actually received the treatment regardless of their initial assignment

COMPARISON

Per-protocol Analysis	As-treated Analysis	Intention-to-treat Analysis
Excludes subjects who did not fulfill the complete protocol	Move subjects between groups after randomization depending on treatment received	Once randomized, always analyzed
Considers compliance	Considers compliance	Disregards compliance
Disturbs the prognostic balance	Disturbs the prognostic balance	Maintains the prognostic balance
Introduces bias	Introduces bias	Minimizes bias
Assessing efficacy under ideal conditions where protocol is followed strictly	Evaluating treatment effect as actually received; useful for safety analyses but prone to bias	Preserving randomization benefits; minimizing bias; reflecting “real-world” adherence issues

INTENTION-TO-TREAT VS PER-PROTOCOL EXAMPLE

- Example: a randomized trial was testing the effect of introducing allergens into the diet of breast-fed children
  - Two groups: early introduction (intervention) vs delayed introduction (control)
  - Primary endpoint was development of food allergy anytime between 1 and 3 years of age

  - ITT analysis with all 1162 participants (567 intervention; 595 control) showed no difference between intervention and control (intervention: 5.6%[32/567] vs control: 7.1% [42/595]). The relative risk (RR) of 0.80 (95% confidence interval [CI], 0.51 to 1.25; P=0.32)

  - PP analysis with 732 participants (208 intervention; 524 control) showed significantly lower frequency of allergy development in intervention compared to control (intervention: 2.4% [5/208] vs control: 7.3% [38/524]. The relative risk (RR) in the early-introduction group was 0.33 (95% CI, 0.13 to 0.83; P=0.01)

REVIEW

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RESEARCH ARTICLE COMMON RED FLAGS

- Method/Study Design:
  - Patient Selection
    - Are the demographics representative of the disease state?
    - Are the demographics in each arm of the study similar? (*allocation concealment*)
    - What are the inclusion/exclusion criteria? *(wide is preferable for improving generalizability)*
    - Do the demographics of the study apply to your patient?
  - Study Design
    - Is the study design and terminology defined?
    - Is the study design appropriate based on population being studied and disease state?
    - Are controls appropriate?
    - Is the type of study appropriate to answer your question?
  - Treatment and Measurement
    - Is the therapy chosen appropriate in terms of dosing, type of patient?
    - Is the duration appropriate to see effect?
    - Were measurements standardized and reliable?
    - Are the outcomes objective and show the effect of the drug?
    - Were primary/secondary objectives defined and appropriate?

RESEARCH ARTICLE COMMON RED FLAGS

- Statistical design
  - Are the statistical tests appropriate for the study design?
  - Is sample size appropriate?
  - Is the power of the study defined?
  - Note: pay special attention to patient population + treatment when looking at a meta-analysis.
- Results
  - Do the results match the outcome objectives of the study?
  - Were all patients accounted for?
    - Intent-to-treat analysis VS. as-per-protocol
  - Are reasons for withdrawal clearly stated?
  - Are the data presented in an unbiased manner?
  - Is the drop-out rate acceptable?
    - Withdrawal < 15% for long-term & 10% for short-term (3 months) studies
  - Does a statistical significance mean clinical significance?

RESEARCH ARTICLE COMMON RED FLAGS

- Discussion/Conclusion
  - Are all conclusions justified by the data presented and factual?
  - Were the strengths and limitations of the study affected by the significance of the results?
  - Note: draw your own conclusions of the study and compare to the conclusions reached by the authors
- References
  - Are the references up to date?
  - Are the most appropriate articles/authors cited?
- Funding
  - Is the funding source independent or sponsored?
    - Pharmaceutical companies may have internal validity issues due to: surrogate outcome measured, dosing regimen not comparable to practice, group studied not reflective of most affected population, publication bias

BIOSTATISTICS

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CONTINGENCY TABLE

- A contingency table shows us the basic interrelations between variables and are fundamental in study design and analysis

	Case	Control
Exposed	A	B
Not Exposed	C	D

ODDS RATIO (OR)

A measure of association between an exposure and an outcome. Tells us the odds that an outcome will occur given a particular exposure compared to the odds that the outcome will occur without the exposure.

	Case	Control
Exposed	A	B
Not exposed	C	D

$$OR = \frac{A/C}{B/D} = \frac{A \times D}{B \times C}$$

OR = 1, exposure has no effect on the odds of the outcome
OR > 1, exposure associated with higher odds of the outcome
OR < 1, exposure associated with lower odds of the outcome

ODDS RATIO (OR) EXAMPLE

We want to see if coffee consumption is linked to heart disease.

	Heart disease	No heart disease
Coffee	40 (A)	60 (B)
No coffee	20 (C)	80 (D)

$$OR = \frac{A/C}{B/D} = \frac{40/20}{60/80} = \frac{2}{0.75} = 2.67$$

Interpretation: The odds of being a coffee drinker are 2.67 x higher among people with heart disease than among people without heart disease.

Key Points:
- OR compares odds → used in case-control studies & logistic regression.
- When the disease is rare (<10%), OR ≈ RR; otherwise, OR can overstate risk.

RELATIVE RISK

Relative Risk (RR): ratio of probability of an outcome in an exposed group to the probability of an outcome in an unexposed group.

	Case	Control
Exposed	A	B
Not exposed	C	D

$IR_e\ (incidence\ rate\ in\ exposed) = EER\ (experimental\ event\ rate)$

$IR_0 = incidence\ rate\ in\ unexposed = CER\ (control\ event\ rate)$

$$\mathrm{RR} = \frac{IR_e/EER}{IR_0/CER} = \frac{(\frac{A}{A+B})}{(\frac{C}{C+D})}$$

- RR = 1, exposure does not affect outcome
- RR > 1, risk of outcome increased by exposure
- RR < 1, risk of outcome decreased by exposure

RELATIVE RISK (RR) EXAMPLE

Example: We want to see if coffee consumption is linked to heart disease.

$$RR = \frac{Risk\ in\ exposed}{Risk\ in\ unexposed} \qquad RR = \frac{(\frac{A}{A+B})}{(\frac{C}{C+D})} = \frac{\frac{40}{40+60}}{\frac{20}{20+80}} = \frac{0.40}{0.20} = 2.0$$

	Heart disease	No heart disease
Coffee	40 (A)	60 (B)
No coffee	20 (C)	80 (D)

Interpretation: coffee drinkers are 2× as likely to develop heart disease as people who do not drink coffee.

Key Points:
- RR compares probabilities → used in cohort studies & RCTs.

RISK REDUCTION/INCREASE

Relative risk reduction (RRR): by how much did the intervention/treatment reduce the risk of disease relative to the untreated group

Absolute risk reduction (ARR): difference between the risk of disease in the untreated and the risk of disease in the treated patients; what is the difference in disease risk [Positive Impact]

Absolute risk increase (ARI): difference between the control event rate (CER) and the experimental event rate (EER); typically used to describe a harmful exposure or intervention [Negative Impact]

	Case	Control
Exposed	A	B
Not exposed	C	D

$$RRR = 1 - RR$$

$$\mathrm{ARR} = \mathrm{CER} - \mathrm{EER} = \frac{\mathrm{C}}{(\mathrm{C}+\mathrm{D})} - \frac{\mathrm{A}}{(\mathrm{A}+\mathrm{B})}$$

$$\mathrm{ARI} = \mathrm{EER} - \mathrm{CER} = \frac{\mathrm{A}}{(\mathrm{A}+\mathrm{B})} - \frac{\mathrm{C}}{(\mathrm{C}+\mathrm{D})}$$

RELATIVE RISK REDUCTION (RRR) EXAMPLE

Example: We want to see if wearing a helmet while tobogganing reduces the risk of head injuries in children.

	Head injury	No head injury
Wearing a helmet while tobogganing	6	83
Not wearing a helmet while tobogganing	19	66

$$RR = \frac{(\frac{A}{A+B})}{(\frac{C}{C+D})} = \frac{6/89}{19/85} = 0.302 \sim 30\%$$

$$RRR = 1 - RR = 1 - 0.302 = 0.698 \sim 69.8\%$$

Interpretation:

- **Children who wear helmets while tobogganing have 30% the risk of head injury compared to children who do not wear helmets.**
- **Children who wear helmets have 69.8% less risk of sustaining a head injury while tobogganing.**

ABSOLUTE RISK REDUCTION (ARR) EXAMPLE

Example: We want to see if wearing a helmet while tobogganing reduces the risk of head injuries in children.

	Head injury	No head injury
Wearing a helmet while tobogganing	6 (A)	83 (B)
Not wearing a helmet while tobogganing	19 (C)	66 (D)

$$ARR = CER - EER = \frac{C}{(C + D)} - \frac{A}{(A + B)} = \frac{19}{19 + 66} - \frac{6}{6 + 83} = 0.156$$

Interpretation:

- **Wearing a helmet lowers a child's risk of sustaining a head injury while tobogganing by 15.6%**

NUMBER NEEDED TO TREAT/HARM

Number needed to treat: the number of subjects you need to treat with exposure to prevent one subject from developing the disease

Number needed to harm: an epidemiological measure that indicates how many persons on average need to be exposed to a risk factor over a specific period to cause harm

$$NNT = \frac{1}{ARR} \qquad NNH = \frac{1}{ARI}$$

Ideally, NNT is low and NNH is high

If calculation leads to a decimal, err on the side of caution

- NNT: round UP. This avoids overestimating benefit
- NNH: round DOWN. This avoids underestimating harm

NUMBER NEEDED TO TREAT (NNT) EXAMPLE

Example: We want to see if wearing a helmet while tobogganing reduces the risk of head injuries in children.

	Head injury	No head injury
Wearing a helmet while tobogganing	6 (A)	83 (B)
Not wearing a helmet while tobogganing	19 (C)	66 (D)

$$NNT = \frac{1}{ARR} = \frac{1}{0.156} = 6.4 \sim 7\ (round\ UP)$$

Interpretation:

- One head injury will be prevented for every 7 children who wear a helmet while tobogganing

HAZARD RATIO (HR)

Measure of how often a particular adverse event occurs in one group compared to another group

Commonly used to report results from RCT in oncology to measure survival at any point in time in a group of patients who have been given a specific treatment compared to a control group given another treatment or a placebo

Hazard Ratio = 1	Both groups experience the same number of events in a period.
Hazard Ratio > 1	The treatment group experiences a higher event probability within any given period than the control group.
Hazard Ratio < 1	The treatment group experiences a lower event probability during a unit of time than the control group.

RRs and ORs assess risk at one point in time, such as the end of a study
In contrast, hazard ratios originate from survival analysis studies that record time-to-event data

SENSITIVITY, SPECIFICITY, PREDICTIVE VALUES

Sensitivity

- Proportion of positive cases that are correctly identified

Specificity

- Proportion of negative cases that are correctly identified

Positive predictive value

- Proportion of patients with positive test who are truly diseased

Negative predictive value

- Proportion of patients with negative test who are truly non-diseased

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\%$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100\%$$

$$\text{Positive predictive value} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\%$$

$$\text{Negative predictive value} = \frac{\text{TN}}{\text{TN} + \text{FN}} \times 100\%$$

	Disease	No disease
Positive Test	True Positive (TP)	False Positive (FP)
Negative Test	False Negative (FN)	True Negative (TN)

SENSITIVITY, SPECIFICITY, PREDICTIVE VALUES EXAMPLE

A new tool is being developed for screening for topical fungal infections. The intention is for patients who test positive to be able to pick up a non-prescription medication from the pharmacy. This removes the need for them to see their family doctor and means they can have access to treatment more quickly. Out of 31 people who were infected, 25 tested positive. There were 46 people who were not infected and out of that, 2 tested positive.

Calculate the sensitivity & specificity.

Step 1: Create the contingency table

Step 2: Calculate the sensitivity + specificity

SENSITIVITY, SPECIFICITY, PREDICTIVE VALUES EXAMPLE

A new tool is being developed for screening for topical fungal infections. The intention is for patients who test positive to be able to pick up a non-prescription medication from the pharmacy. This removes the need for them to see their family doctor and means they can have access to treatment more quickly. Out of 31 people who were infected, 25 tested positive. There were 46 people who were not infected and out of that, 2 tested positive.

Calculate the positive predictive value (PPV) and the negative predictive value (NPV).

SENSITIVITY, SPECIFICITY, PREDICTIVE VALUES EXAMPLE

Step 1: Create the contingency table

	Fungal infection	No fungal infection
Positive test	25 (TP)	2 (FP)
Negative test	6 (FN)	44 (TN)

Step 2: Calculate the sensitivity + specificity

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\%$$

$$\text{Sensitivity} = \frac{25}{25 + 6} \times 100\% = 81\%$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100\%$$

$$\text{Specificity} = \frac{44}{44 + 2} \times 100\% = 96\%$$

Step 3: Calculate the positive predictive value (PPV) and the negative predictive value (NPV).

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% = \frac{25}{25 + 2} \times 100\% = 93\%$$

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \times 100\% = \frac{44}{44 + 6} \times 100\% = 88\%$$

ATTRIBUTABLE RISK (AR)

Attributable risk (AR) is the difference between the incidence rate in the exposed group and the incidence rate in the unexposed group.

	Case	Control
Exposed	A	B
Not exposed	C	D

$$AR = IR_e - IR_0 = \frac{A}{A+B} - \frac{C}{C+D}$$

ATTRIBUTABLE RISK EXAMPLE

Suppose that you conducted a cohort study to measure the correlation between smoking and lung cancer. 20% of your sample group were active smokers. You followed your cohort for 20 years, and of those who smoked, 59% developed lung cancer. Of those who did not smoke, only 0.125% developed lung cancer.

What is the attributable risk of lung cancer to smoking?

$$AR = IR_e - IR_0 = 0.59 - 0.00125 = 0.589$$

Interpretation: The absolute difference in risk of developing lung cancer between people who smoke and people who do not smoke is 58.9%.

ATTRIBUTABLE FRACTION (AF)

Attributable fraction (AF) is the proportion of the outcome *in the sample group* that can be attributed to the exposure.
It is calculated by finding the difference between the incidence in the exposed and the incidence in the unexposed and dividing by the incidence in the exposed:

$$AF = \frac{(IR_e - IR_0)}{IR_e} \times 100\%$$

$$AF = \frac{\frac{A}{A+B} - \frac{C}{C+D}}{\frac{A}{A+B}} \times 100\%$$

Alternatively, the following equations may be used:

$$AF = \frac{OR - 1}{OR} \times 100\%$$

$$AF = \frac{RR - 1}{RR} \times 100\%$$

ATTRIBUTABLE FRACTION (AF) EXAMPLE

Suppose that you conducted a cohort study to measure the correlation between smoking and lung cancer. 20% of your sample group were active smokers. You followed your cohort for 20 years, and of those who smoked, 59% developed lung cancer. Of those who did not smoke, only 0.125% developed lung cancer.

What is the attributable fraction of lung cancer to smoking in this cohort?

$$AF = \frac{IR_e - IR_0}{IR_e} \times 100\% = \frac{0.59 - 0.00125}{0.59} \times 100\% = 99.8\%$$

Interpretation: Among smokers in this cohort, 99.8% of lung cancer cases can be attributed to smoking.

POPULATION ATTRIBUTABLE RISK (PAR)

Population attributable risk (PAR) is the difference between the incidence rate in the total population and the incidence rate in the unexposed.

PAR = incidence rate in the total population– incidence rate in the unexposed group

$$PAR = IR_{total\ population} - IR_0$$

Alternatively, it can be calculated using the following formula, where P_e is the prevalence of the exposure in the population.

$$PAR = AR \times P_e$$

POPULATION ATTRIBUTABLE RISK (PAR) EXAMPLE

You are studying the correlation between NSAID use and peptic ulcer disease. You know that at any given time, approximately 0.1% of the Canadian population has active peptic ulcer disease. The risk of peptic ulcer disease is 0.04% in patients who do not use NSAIDs.

What is the population attributable risk (PAR)?

$$PAR = IR_{pop} - IR_0 = 0.1\% - 0.04\% = 0.06\%$$

Interpretation: 0.06% of the Canadian population have active peptic ulcer disease specifically due to NSAID use

POPULATION ATTRIBUTABLE FRACTION (PAF)

Population Attributable Fraction (PAF) is the proportion of the outcome *in the total population* that can be attributed to the exposure. It is calculated using the following formula:

$$PAF = \frac{IR_{total\,population} - IR_0}{IR_{total\,population}}$$

$$PAF = \frac{PAR}{IR_{total\,population}}$$

Alternatively, he following formulas may be used, where P_e is the prevalence of exposure in the population.

$$PAF = \frac{P_e \times (OR - 1)}{1 + P_e \times (OR - 1)} \times 100\%$$

$$PAF = \frac{P_e \times (RR - 1)}{1 + P_e \times (RR - 1)} \times 100\%$$

POPULATION ATTRIBUTABLE FRACTION (PAF) EXAMPLE

You are studying the correlation between NSAID use and peptic ulcer disease. You know that at any given time, approximately 0.1% of the Canadian population has active peptic ulcer disease. The risk of peptic ulcer disease is 0.04% in patients who do not use NSAIDs.

What is the population attributable fraction (PAF)?

$$PAF = \frac{IR_{pop} - IR_0}{IR_{pop}} \times 100\% = \frac{0.1\% - 0.04\%}{0.1\%} \times 100\% = 60\%$$

Interpretation: 60% of peptic ulcer disease cases in the Canadian population can be attributed to NSAID use.

HYPOTHESIS

A well worked up hypothesis is half the answer to the research question!

Hypothesis

- Educated guess; proposed explanation for a phenomenon that can be tested
- Must be based on a good research question (PICO)
- Should be simple, specific and stated in advance

Example: Positive family history of schizophrenia increases the risk of developing the condition in first-degree relatives

- Simple → family history; risk of disease
- Specific → no room for ambiguity

TYPES OF HYPOTHESIS

Null hypothesis (H_0): general statement or default position

The null hypothesis is the basis for the statistical test

No difference between two phenomena

Alternative hypothesis(H_A): proposed statement

There is a difference between two phenomena

Using hypothesis testing, you can make decisions about whether your data support or refute your research predictions with null and alternative hypotheses.

Either reject or fail to reject the H_0

Since these decisions are based on probabilities,
there is always a risk of arriving at the wrong conclusion

TYPE I/TYPE II ERRORS

- Type I (α)
  - Conclude there is a difference between the groups when there is no actual difference between them
  - Rejecting the null hypothesis when it is true
- Type II (β)
  - Conclude that there is no difference between the groups when there is an actual difference between them
  - Failing to reject the null hypothesis when it's actually false
  - Commonly due to small sample size

DECISION	H_0 True	H_0 False
Accept H_0	Correct decision	Type II error (β)
Reject H_0	Type I error (α)	Correct decision

If p is low → Reject the H_0

TYPE I ERROR

Rejecting the null hypothesis when it's actually true

It means concluding that results are statistically significant when, in reality, they came about purely by chance or because of unrelated factors

The risk of committing this error is the significance level (alpha or α) you choose

- Alpha (α) → a value that you set at the beginning of your study to assess the statistical probability of obtaining your results (p value)
    - α = 0.05 is most common

- If p-value of your study is lower than 0.05 → results are statistically significant and consistent with the alternative hypothesis
- If p-value of your study is higher than 0.05 → results are statistically nonsignificant

If p is low → Reject the H_0

The p-value is the evidence against a null hypothesis
The smaller the p-value, the stronger the evidence that you should reject the null hypothesis.

TYPE II ERROR

Failing to reject the null hypothesis when it's actually false

In reality, your study may not have had enough statistical power to detect an effect of a certain size
Power is the extent to which a test can correctly detect a real effect when there is one
A power level of 80% or higher is usually considered acceptable.

Power can be affected by:

- Significance level: increasing the significance level increases power
- Sample size – the smaller or more unbalanced the group, the lower the power of the study; the larger the sample size, the better the power
- Effect size – the larger the effect, the higher the power since it is easier to find large effect

To (indirectly) reduce the risk of a Type II error, you can increase the sample size or the significance level

UNDERSTANDING THE P-VALUE

Question 1. A recent discovery for a new medication for the treatment of dyslipidemia was found and called Drug X. Is Drug X an effective treatment option for patients with dyslipidemia?

Hypothesis: Drug X is more likely to reduce lipid levels in patients with dyslipidemia than standard statin therapy

Which of the following statements indicates the highest statistical significance?

a. In a trial, participants who were prescribed Drug X (n=150) were 5 times more likely to have decreased lipid levels than those who were prescribed atorvastatin (n=150), $p < 0.02$.
b. In a trial, participants who were prescribed Drug X (n=150) were 5 times more likely to have decreased lipid levels than those who were prescribed atorvastatin (n=150), $p < 0.08$.
c. In a trial, participants who were prescribed Drug X (n=150) were 5 times more likely to have decreased lipid levels than those who were prescribed atorvastatin (n=150), $p < 0.001$.

Question 2. In each of the above statements, are we rejecting or accepting the null hypothesis?

PRACTICE PROBLEMS

Qualifying Exam Preparatory Course

PharmAchieve

PRACTICE QUESTION 1

A new herb is being investigated for its potential use in the management of insomnia. Out of a total of 151 participants, 48 patients were given the placebo, of which 15 experienced insomnia. There were 75 patients who took the herb but did not experience insomnia. Calculate the relative risk, relative risk reduction, absolute risk reduction and number needed to treat.

Step 1: Create the contingency table

Step 2: Calculate the Relative Risk

Step 3: Calculate the Relative Risk Reduction

PRACTICE QUESTION 1

Step 4: Calculate the following:

Absolute Risk Reduction:

Number Needed to Treat:

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CHANGE LOG

August 2025

- Major formatting changes made throughout
- Re-ordered slides to improve the flow of the lecture
- Multiple edits/additions made to slides
- Slide 4: Elaborated on the table of contents
- Slide 17: Added healthy user bias, recall bias & volunteer bias
- Slide 64-69: Removed the answers to make it more interactive
- Added slide example for RR vs OD

November 2025

- Font size increased throughout
- Slide 27: Added definition of meta-analysis/systematic reviews
- Slide 58: Added another AR equation
- Slide 71: Updated references

February 2026

- Formatting updated throughout
- Slide 2: Updated NAPRA competencies
- Slides 35-36: Modified definition of case-control studies and cohort studies for more clarity
- Slides 56-57: Added example of relative risk reduction (RRR) & absolute risk reduction (ARR)
- Slide 59: Added number needed to treat (NNT) example
- Slides 62-64: Relocated sensitivity, specificity, predictive values example for better clarity/flow
- Slides 66-73: Added definitions, equations and examples for AR, AF, PAR & PAF